

Latest insights into the risk of cancer in diabetes

A growing body of evidence from observational studies and meta-analyses of the data suggest that diabetes mellitus is associated with an increased risk of cancer. Meta-analyses have shown that diabetes increases the risks of total cancer, and of site-specific cancers of the breast, endometrium, bladder, liver, colorectum and pancreas, and that it decreases the risk of prostate cancer. Insulin resistance and secondary hyperinsulinemia is the most frequently proposed hypothesis, and hyperglycemia itself might promote carcinogenesis. In addition to several facets of lifestyle including obesity, smoking and lack of exercise, treatment for diabetes might affect the risk of cancer. For instance, metformin, an insulin sensitizer, reportedly has a potential anticancer effect. In light of the exploding global epidemic of diabetes, even a modest increase in the cancer risk will translate into a substantial socioeconomic burden. The current insights underscore the need for clinical attention and better-designed studies of the complex interactions between diabetes and cancer.